

HDD SELECTOR SWITCH

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Using the switch-mode power supply (SMPS) of your personal computer, this add-on circuit lets you switch between three hard disk drives (HDDs) and also ensure that nobody else can open your protected HDD. It is quite useful for protection from hacking and spying.

Fig. 1 shows the block diagram of the HDD selector switch. You can select between three different HDDs such as main, net and backup within a certain time period (say, 20 seconds) after the PC is switched on. After the

predetermined time, no HDD can be opened without restarting your PC.

As shown in Fig. 2, the HDD selector switch is built around IC CD4093 (IC1) and a few discrete components. IC CD4093 contains four 2-input Schmitt NAND gates. Pin 1 of gate N1 is connected to the positive supply via capacitor C1 and diode D1, and to the ground (GND) via resistor R1. Pin 2 of gate N1 is pulled low by resistor R2. Master lock switch is also connected between +12V and pin 2. Pin 3 of gate N1 is connected to inputs pins 8-9 and 12-13 of gates N3 and N4, respectively, and the outputs of these gates are fed to gate N2. A bicolour LED (LED1) is

connected to output pins of N3 and N4 and input pins of N2 for glowing green and output pin 4 of N2 for glowing red. The junction of pins 5, 6, 10 and 11 is connected to the common point of switches S2 through S4.

Switches S2 through S4 are used to trigger the gates of SCR1 through SCR3 (for 12V supply) and

SCR4 through SCR6 (for 5V supply) through diodes and resistors. For indication of the selected HDD, three green LEDs (LED2 through LED4) are connected between the cathodes of SCR1 through SCR3 and grounds. You will get 12V at the cathodes of SCR1 through SCR3 and 5V at the cathodes of SCR4 through SCR6 when the respective switches (S2 through S4) are pressed.

Working of the circuit is simple. Initially, when the SMPS is 'on' and lock switch S1 is closed, bicolour LED1 glows green for around 20 seconds. In this state, any HDD can be connected to the power supply through switches S2 through S4. For example, if you want to activate the net HDD, press switch S3 momentarily within 20 seconds to get 5V and 12V power supplies

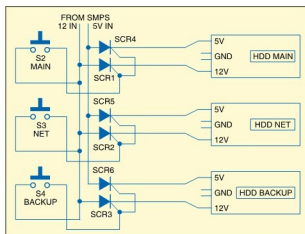


Fig. 1: Block diagram of hard disk drive selector (HDD)

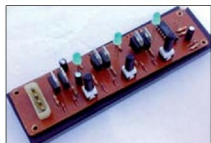


Fig. 3: Author's prototype

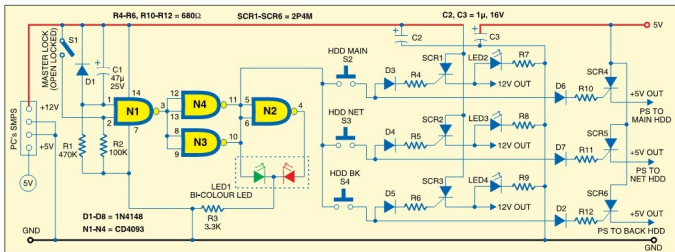


Fig. 2: Circuit of HDD selector switch